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# POCUS-Emergency

Intelligent Multimodal Assistant for Clinical Ultrasound  
and Decision Support in Emergencies

*A multimodal decision-support system combining patient information, three ultrasound modules and a clinical reasoning layer — with an explicit account of what it does not know.*

4

perception models

339

automated tests

37

safety properties

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# The problem

*Emergency assessment is multimodal, and the modalities arrive apart*

1

## Time pressure

Decisions are made before all the information is available.

2

## Scattered evidence

Vitals, complaint, laboratory values and imaging arrive separately.

3

## Operator dependence

POCUS adds real information at the bedside — its interpretation varies with the operator.

4

## Integration

Combining these under time pressure, while tracking what is still unknown, is where error enters.

### The gap

**No system combines  
imaging with the rest  
of the emergency record  
while keeping an honest  
account of its own  
reliability.**

*That gap is the subject of this work.*

**The objective is assistance, not replacement. The physician remains responsible for the clinical decision.**

# The proposed solution

*One workflow, several sources of evidence, one decision*

## Centralised evidence

Triage, three POCUS modules and the clinical record on one screen.

## Explicit uncertainty

Every finding carries its reliability, and absent evidence is labelled absent — never normal.

## Structured reasoning

The assessment is assembled from enumerated evidence, not narrated freely.

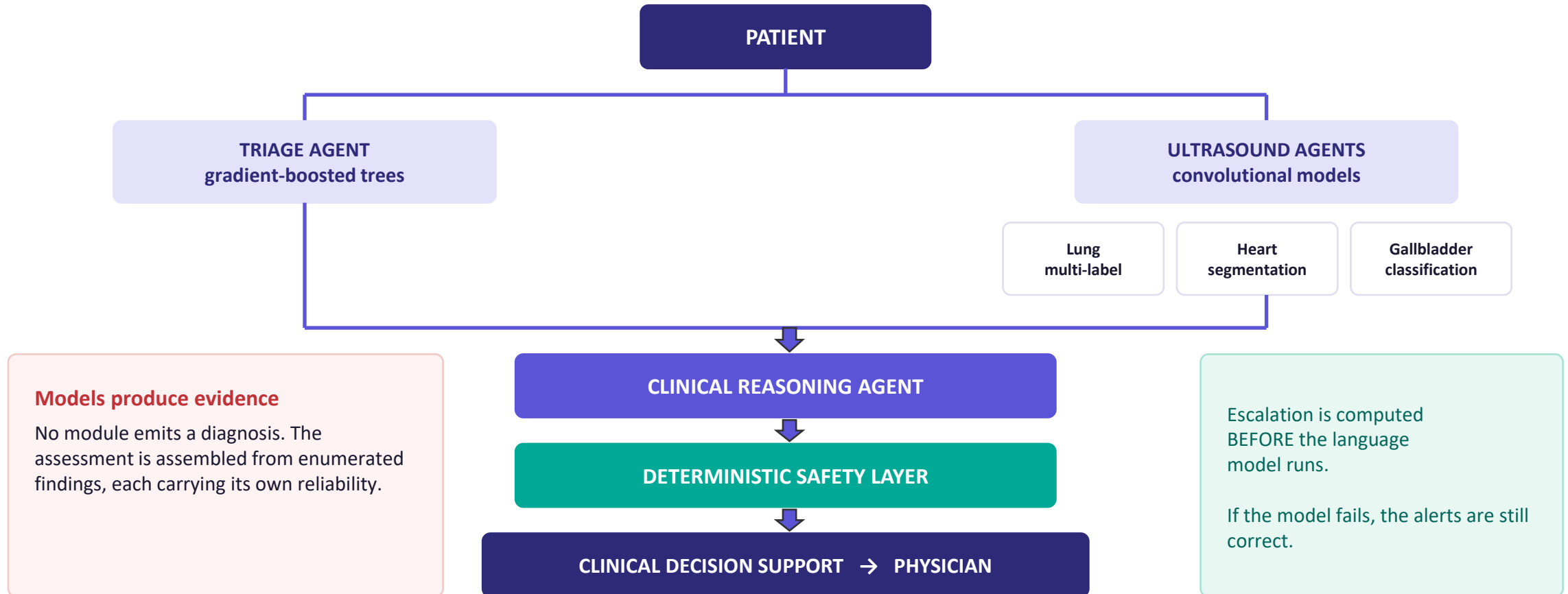
## Persistent record

Each physician keeps their own patients, examinations and images.

**Delivered as a working web platform with accounts, a stored patient history, containerised deployment and an automated test suite.**

# Technical architecture

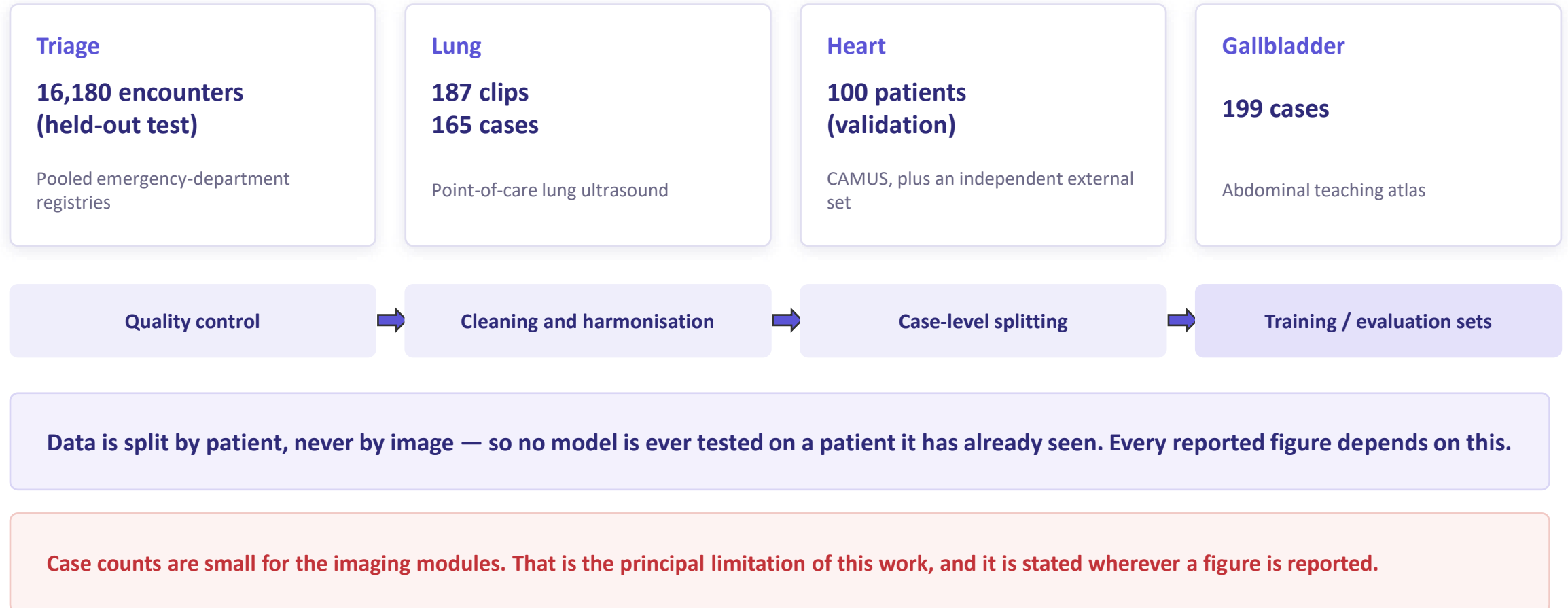
CRISP-DM · Modeling



# Data sources and preparation

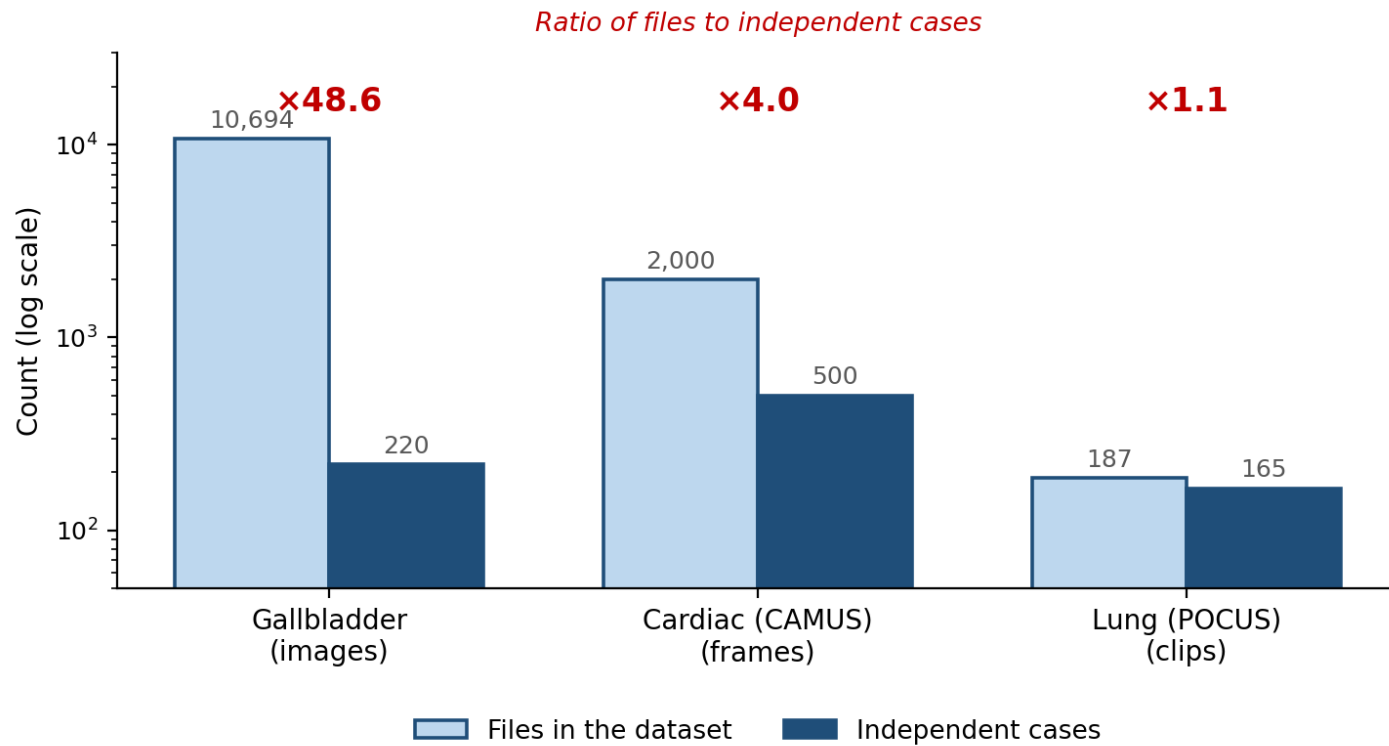
CRISP-DM · Phases 2–3

*Four independent public sources, prepared under one rule*



# Why the split is by case, not by image

*The same patient appears many times over — counting files would count them as many patients*



## What the ratio means

The gallbladder set holds 10,694 images from 220 patients — on average 48 images of the same person.

Split those files at random and the same patient lands in both training and test. The model recognises the patient, the score rises, and nothing has been learned about a new one.

**Every figure in this presentation is computed on splits made by case.**

# Module 1 — Triage

Modeling

## Clinical task

Grade the urgency of a patient from information available at the bedside.

## Data

Pooled emergency-department registries. Disposition fields removed after they were found to leak the outcome into the input.

## Approach

Gradient-boosted trees on 21 prepared inputs — 10 entered by the clinician, 11 derived (shock index, pulse pressure, red-flag indicators). Probabilities calibrated.

## Output

A suggested urgency tier with its probability — presented to the clinician, who confirms or changes it. The suggestion is never applied on its own.

**The deployed model was retrained on exactly the fields the interface collects, so its reported accuracy is accuracy a bedside prediction can actually reach.**

# Results — Triage

CRISP-DM · Evaluation

*Held-out test set, 16,180 encounters*

**63.0%**

Accuracy

**56.7%**

Macro F1

**77.3%**

High-urgency recall

**1.36%**

Dangerous miss rate

## Two models, and why the weaker one is deployed

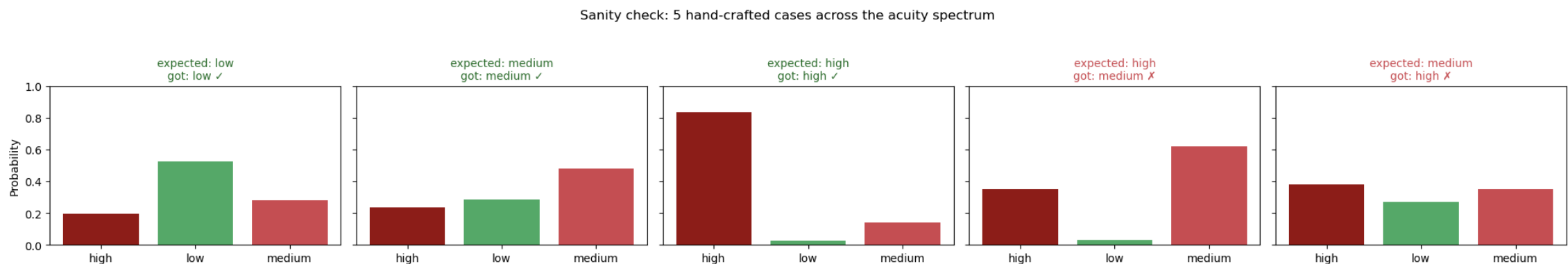
The reference model also sees hospital registration fields and reaches 68.0% accuracy. The deployed model uses only what a clinician types at the bedside: 63.0% accuracy, and a LOWER dangerous-miss rate — 1.36% against 1.55%. On the error that matters, the deployable model is the better one.

**A dangerous miss is a genuinely high-acuity patient graded low. These are retrospective figures on registry data: a decision-support suggestion, not a validated triage instrument.**

*Model selection: gradient-boosted trees were compared against logistic regression, random forest, a multilayer perceptron and CatBoost on identical features and splits. None beat them — the ceiling is a property of the data.*

# Triage: a behavioural check on constructed cases

Five cases written by hand to span the acuity spectrum



| Constructed case                                      | Expected | Predicted     | Conf. |   |
|-------------------------------------------------------|----------|---------------|-------|---|
| Healthy young adult, minor ankle sprain, walk-in      | low      | low           | 52.2% | ✓ |
| Middle-aged, moderate abdominal pain, stable vitals   | medium   | medium        | 47.8% | ✓ |
| Elderly, tachycardic, hypotensive, hypoxic, ambulance | high     | high          | 83.2% | ✓ |
| Elderly, ambulance, severe chest pain, tachycardic    | high     | <i>medium</i> | 61.9% | ✗ |
| Young adult, high fever and tachycardic, walk-in      | medium   | <i>high</i>   | 37.9% | ✗ |

Five constructed cases support no statistical claim. The purpose is to see whether the model moves in the direction a clinician would expect — which an aggregate accuracy cannot show.

# Module 2 — Lung

Modeling

## Clinical task

Detect four findings on point-of-care lung ultrasound: B-lines, consolidation, pleural effusion, pleural thickening.

## Data

187 clips from 165 cases. Frames sampled per clip; clips of one case never split across folds.

## Approach

EfficientNet-B0 backbone with a multi-label head — findings co-occur, so they must not compete for probability mass. Three-fold grouped cross-validation, per-finding thresholds.

## Output

Per-finding detection with a confidence, reduced across frames by median so one blurred frame cannot carry the result.

**Pneumothorax is deliberately NOT modelled: it is a motion sign, invisible in a single static frame, and the dataset contains no positive cases.**

# Results — Lung

Pooled out-of-fold, 187 clips from 165 cases

| Finding            | AUROC | Sensitivity | Positive cases |
|--------------------|-------|-------------|----------------|
| B-lines            | 0.830 | 84.1%       | 63             |
| Consolidation      | 0.832 | 60.0%       | 60             |
| Pleural effusion   | 0.718 | 75.0%       | 24             |
| Pleural thickening | 0.773 | 65.2%       | 46             |

Macro AUROC 0.788

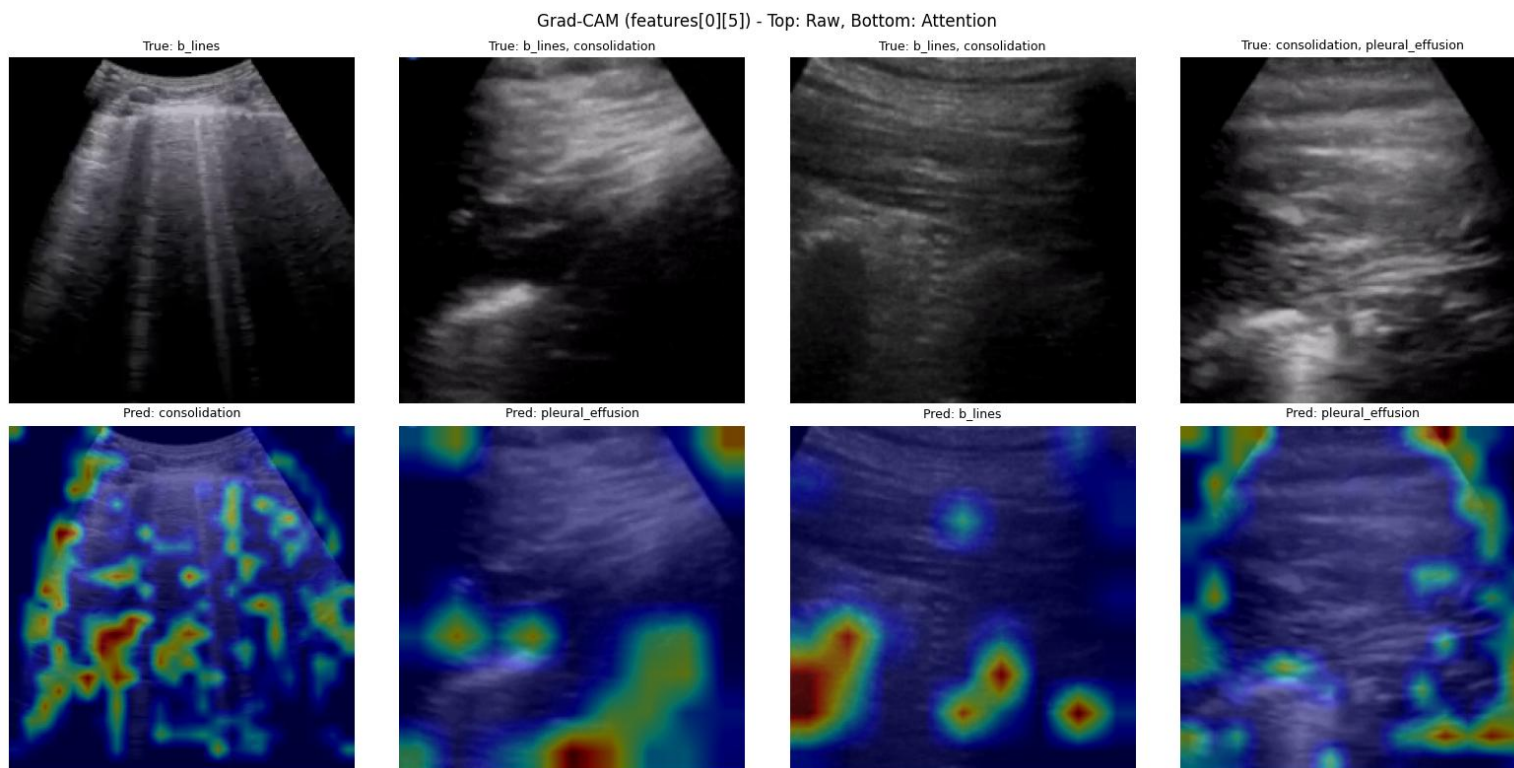
Pleural effusion rests on 24 positive cases, and per-fold AUROC for B-lines ranges from 0.536 to 0.830. A pooled figure conceals that spread, which is why the spread is reported.

Pneumothorax is not modelled and therefore cannot be excluded by this system. It is reported as not assessed — never as absent.

Calibration was not applied to this module: a held-out calibration split would leave about three positive effusion cases, and a curve fitted on three points is not a calibration. The interface states that the confidence is uncalibrated.

# What the lung model attends to

*Grad-CAM on cases the model got WRONG — a selection of successes would prove nothing*



## Why the failures

A viewer who already knows where the pathology lies will read any attention map as pointing at it. Maps of correct predictions are therefore not evidence.

Shown here are cases the model misclassified, so attention on a correct and an incorrect prediction can be compared. That comparison is the only one from which anything follows without expert annotation.

*Aggregate statistics — centroid spread, area above half-maximum, centre-bias — were computed over the full sets.*

# Module 3 — Heart

Modeling

## Clinical task

Segment the left ventricle and derive an ejection-fraction band from the segmentation.

## Data

CAMUS, 100 validation patients, plus an independent external collection used for validation only and never for training.

## Approach

U-Net with an EfficientNet-B0 encoder, four output classes. Ejection fraction computed from end-diastolic and end-systolic areas, with a geometric volume correction.

## Output

Left-ventricle segmentation and an ejection-fraction band — severe, reduced or normal — with the value flagged when it falls close to a cutoff.

**Segmentation quality and clinical output are reported separately. The band is what a clinician acts on, so the band is the module's result.**

# Results — Heart

CRISP-DM · Evaluation

*Segmentation is strong; the derived clinical band is weaker, and that is the figure reported*

**0.928**

LV Dice, in-domain  
(nnU-Net: 0.93)

**0.534**

LV Dice, external

**63.4%**

EF-band balanced acc.  
(chance 33.3%)

**6.9 pp**

EF mean absolute error  
(was 12.2 pp)

**Ejection fraction here is a geometric estimate derived from segmentation — a proxy, not a definitive echocardiographic measurement. Volume correction halved its error.**

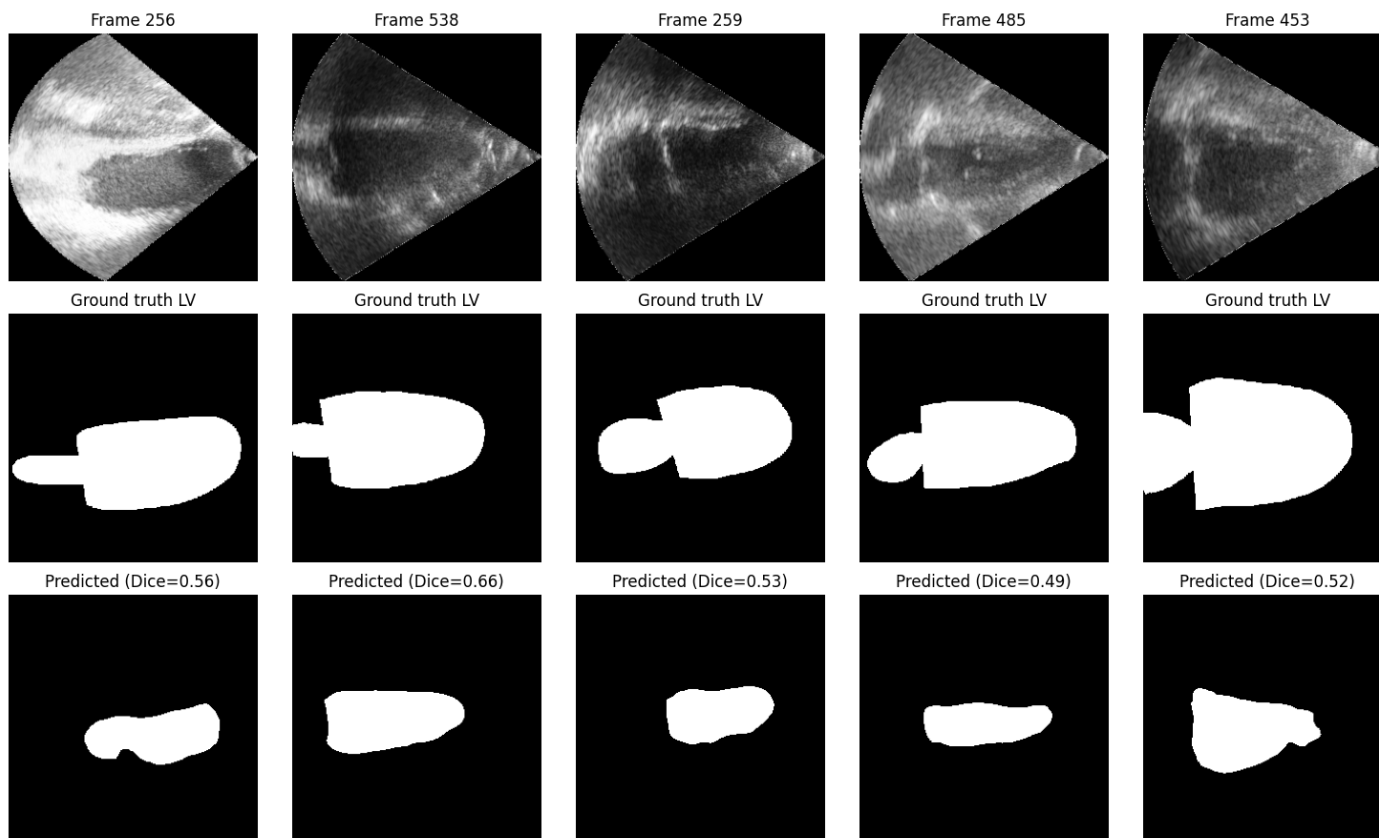
## **Values near a band boundary are flagged**

A 6.9-point error is enough to move a patient across a cutoff. Where an estimate falls within that margin of a threshold, the system states the value could lie on either side. The margin is the module's own measured error, not a round number chosen for convenience.

*The external drop was first attributed to a labelling-convention mismatch. Connected-component analysis over the full external set disproved that hypothesis, and the reversal is reported rather than dropped.*

# Where the cardiac model fails

*Five frames from the external collection: the image, the reference contour, and what the model produced*



## A consistent failure, not a random one

The predicted region is systematically SMALLER than the reference, and loses the apex. It is not noise: the model finds the ventricle and under-segments it.

Dice on these frames runs 0.49 to 0.66, against 0.928 in-domain. Under-segmenting the ventricle biases the derived ejection fraction, which is why the band and not the Dice is reported as the module's clinical result.

*Showing the failure is the point: an external gap named but never displayed is a gap a reader cannot judge.*

# Module 4 — Gallbladder

Modeling

## Clinical task

Identify the principal abnormality in a gallbladder study, among five reported groups.

## Data

199 cases from an abdominal teaching atlas. One class was excluded after per-case inspection showed its contents did not match its label.

## Approach

EfficientNet-B0 trained on eight fine-grained classes, reported as five by summing group probabilities. Temperature scaling for calibration.

## Output

One reported finding with a calibrated confidence. Groups are mutually exclusive: classes considered and rejected are not passed on as evidence.

**The atlas contains no healthy class. This module can say which abnormality is most consistent with the image — it cannot say a gallbladder is normal.**

# Results — Gallbladder

CRISP-DM · Evaluation

*Five reported groups, held-out evaluation*

**59.7%**

Balanced accuracy  
(chance 20.0%)

**0.024**

Calibration error  
after temperature scaling

**199**

Cases

## Why the taxonomy matters more than the accuracy

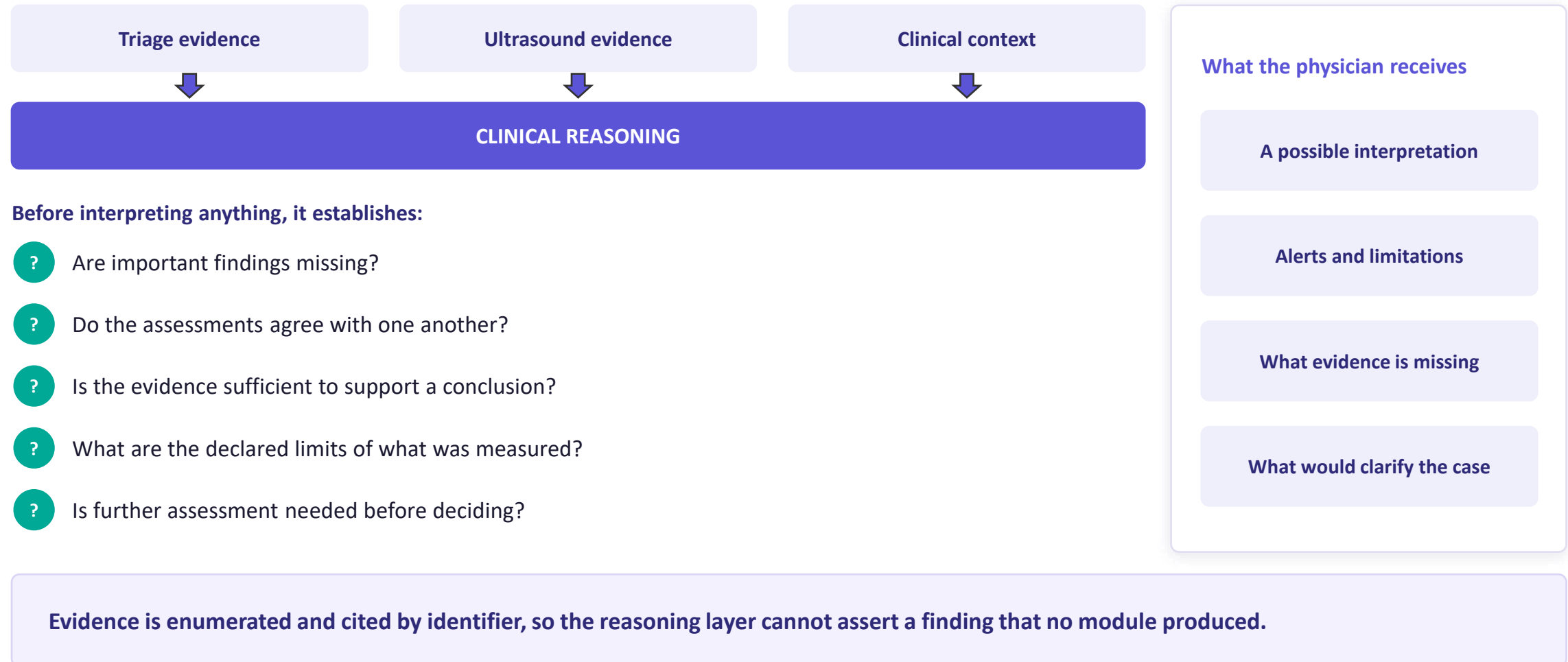
82% of the model's confusions cross a reported group boundary. A coarser three-category grouping was tested and rejected on that evidence: if the groups captured a distinction the model could see, its confusions would concentrate within them. They do not.

**No healthy class. The model reports which abnormality is most consistent with the image; it cannot report the absence of disease, and the interface says so beside every result.**

*A ninth folder present in the original release was excluded after sampling one frame per case across the whole set. A first inspection of a handful of frames had judged it consistent — frames from one case are not a sample of a class.*

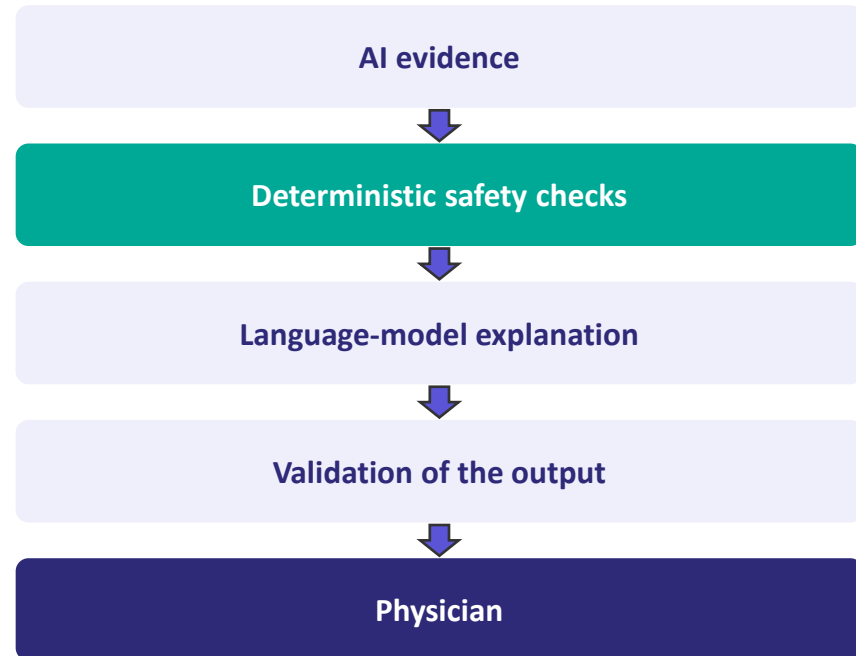
# The Clinical Reasoning Agent

*How the branches are combined*



Modeling

# Safety architecture



Computed before any  
model runs:

Missing evidence  
Disagreement  
Severity  
Uncertainty

## Four states a finding can occupy

 **detected**  
a positive finding

 **unreliable**  
detected, weak evidence

 **screened, not detected**  
looked for, not found

 **not assessed**  
no model covers this

**Absence of evidence is never recorded as evidence of absence. This is enforced by the data structure, not by a convention someone must remember.**

# Global validation

CRISP-DM · Evaluation

*What was verified, and how*

**339**

Automated tests  
all passing

**37**

Safety properties  
each reported separately

**150**

Regression checks  
over 10 full cases

**6 / 6**

Automated checks  
tests, lint, security

## Grouped by property, not by count

- Conflict detection — 14 / 14
- Absent is not normal — 24 / 24
- Hallucination rejection — 25 / 25
- Access control — 17 / 17

*"339 passed" is not evidence. "Conflict detection 14/14" is.*

## Beyond model metrics

- Generalisation tested on an independent set where one existed
- Calibration measured per module, and refused where data was insufficient
- Whole encounters run end to end, intake through report
- Individual real studies inspected by hand

**Seven well-motivated changes were tested and rejected on measurement rather than preference — each reported with its outcome.**

# Technical deployment

CRISP-DM · Deployment

*A running application, not a notebook*

1

## Application

FastAPI and uvicorn serving a generated clinician interface in English and French

2

## Containerised

Docker image running as a non-root user, with a health check that reports whether each model actually loaded

3

## Accounts and storage

Per-physician accounts, hashed passwords, server-side sessions, and a stored patient record

4

## Automated pipeline

Every check runs on each push: tests, regression, lint, code and dependency security

**Technical deployment and clinical deployment are different claims. The first is demonstrated here; the second — installation in a hospital and prospective validation — has not been attempted.**

# The clinician interface

The interface is titled "POCUS-Emergency Clinical Copilot". It features a sidebar on the left with navigation options: Home, Patient workup, Diagnosis, Patient record, Assessment (selected), Alerts (7), Clinical assistant, Timeline, Report, and History. The main content area is titled "Clinical assessment" for "Patient A · 78 · F · shortness of breath". A "Generate clinical report" button is in the top right. A red alert box at the top states: "HIGH CLINICAL PRIORITY. Acute dyspnoea presentation assessed at HIGH severity. Point-of-care ultrasound detected b lines (lung, 0.82), pleural effusion (lung, 0.58). NOT ASSESSED: heart — expected for this presentation and never examined. 3 critical alert(s): spo2 88.0 % is below the configured critical bound of 90.0 %; a positive imaging finding with troponin, lactate never measured; expected but never assessed: heart — a gap in the record, not a negative result. ESCALATED — 2 trigger(s): route simulation. Most informative next: heart ultrasound, troponin, lactate." A "Review alerts" button is next to it. Below the alert, the interface is divided into three columns. The left column, "What the assistant sees", lists symptoms (shortness of breath), vital signs (hr 118 bpm, sbp 96 mmHg, rr 26 /min, spo2 88 %, temp 37.4 C, dbp 58 mmHg, pain 4 /10), POCUS findings (b lines — Detected (0.82), pleural effusion — Detected — weak evidence (0.58), consolidation — Screened, not detected (0.05), pneumothorax — NOT assessed — not modelled (—)), and laboratory results (troponin, bnp, d\_dimer, lactate, crp, wbc, creatinine, ph — all "Not measured"). The middle column, "Differential assessment", states: "Offered for consideration. This is not a diagnosis. No differential was generated. Producing one requires a 4.9 GB language model on a GPU, which is not loaded in this deployment. Everything else on screen was computed here." The right column, "What would help clarify this case?", lists tests not performed: heart ultrasound, troponin, lactate, bnp, d\_dimer, and crp, each with a brief explanation. A "CURRENT PATIENT" box at the bottom left shows "Patient A" with "78 · F · shortness of breath" and a "HIGH" status. A bottom bar shows "AR Perception modules" for lung, heart, and gallbladder.

**POCUS-Emergency Clinical Copilot**

Search patients, findings, reports...

Assistant ready Français + New assessment

## Clinical assessment

Patient A · 78 · F · shortness of breath

**HIGH CLINICAL PRIORITY**  
Acute dyspnoea presentation assessed at HIGH severity. Point-of-care ultrasound detected b lines (lung, 0.82), pleural effusion (lung, 0.58). NOT ASSESSED: heart — expected for this presentation and never examined. 3 critical alert(s): spo2 88.0 % is below the configured critical bound of 90.0 %; a positive imaging finding with troponin, lactate never measured; expected but never assessed: heart — a gap in the record, not a negative result. ESCALATED — 2 trigger(s): route simulation. Most informative next: heart ultrasound, troponin, lactate.

**What the assistant sees**  
The information available for this patient.

**SYMPTOMS**  
shortness of breath

**VITAL SIGNS**  
hr 118 bpm  
sbp 96 mmHg  
rr 26 /min  
spo2 88 %  
temp 37.4 C  
dbp 58 mmHg  
pain 4 /10

**POCUS**  
b lines — Detected (0.82)  
pleural effusion — Detected — weak evidence (0.58)  
consolidation — Screened, not detected (0.05)  
pneumothorax — NOT assessed — not modelled (—)

**LABORATORY**  
troponin — Not measured  
bnp — Not measured  
d\_dimer — Not measured  
lactate — Not measured  
crp — Not measured  
wbc — Not measured  
creatinine — Not measured  
ph — Not measured

**Differential assessment**  
Offered for consideration. This is not a diagnosis.  
No differential was generated. Producing one requires a 4.9 GB language model on a GPU, which is not loaded in this deployment. Everything else on screen was computed here.

**What would help clarify this case?**  
These tests were not performed. Their absence does not mean they are normal.

- heart ultrasound**  
expected for this presentation and never assessed; an unperformed scan is missing information, not a negative result
- troponin**  
named by an escalation trigger as a key value that was never measured
- lactate**  
named by an escalation trigger as a key value that was never measured
- bnp**  
never measured; absent rather than normal
- d\_dimer**  
never measured; absent rather than normal
- crp**  
never measured; absent rather than normal

**CURRENT PATIENT**  
Patient A  
78 · F · shortness of breath  
• HIGH

**AR Perception modules**  
• lung • heart • gallbladder

## On one screen

- Patient information
- Triage suggestion
- Ultrasound findings
- What was not measured
- Alerts and limitations
- What would clarify the case

*Every value on the screen is computed by the pipeline. Where the system has nothing real to put in a slot, the slot says so.*

*Served in English and in French — the French text is composed from the same computed fields, so a French screen cannot state something the English record does not.*

# What this could bring to an emergency department



## Centralised evidence

Triage, imaging and laboratory values assembled in one place rather than read separately.



## Faster access to findings

A POCUS study is read and filed against the patient as it is acquired.



## Explicit uncertainty

Every finding carries its reliability, and what was not measured is named.



## Structured reasoning support

A written assessment with its alerts, its gaps and what would clarify the case.



## Persistent history

Each physician's patients, examinations and images kept across visits.



## Extensible

Further organs and findings slot into the same contract and the same safety layer.

**The value proposition is the workflow, not any single model. No individual module reaches the accuracy a standalone diagnostic tool would require.**

# Limitations and roadmap

*Stated plainly — a system that hides these is not usable clinically*

## Current limitations

- Small datasets for several ultrasound findings
- External generalisation remains difficult (LV Dice 0.928 → 0.534)
- Ejection fraction is a geometric proxy
- Pneumothorax is not assessed at all
- The gallbladder model has no healthy class
- No prospective clinical validation

## Roadmap

- Larger and more diverse clinical datasets
- External and prospective validation
- Improved robustness of the ultrasound modules
- Evaluation with emergency physicians
- Additional organs within the same safety contract
- Integration into a real clinical workflow

**The system is built to accept validated data later: versioned by content hash, retrained, evaluated against fixed criteria, and approved by a named person before any model reaches a patient.**

# What it can tell you — and what it cannot

## It can

assemble the available evidence, grade urgency, read three organ systems, and state what is missing.

## It cannot

exclude a pneumothorax, confirm a normal gallbladder, or replace an echocardiographic measurement.

## It never

makes the clinical decision. That remains yours.

339 automated tests · 37 safety properties · 150 regression checks · 4 perception models · a working platform